**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 104 · Lesson 104 · Magnetic, Optical and Flash Storage

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does non-volatile mean?

Q2. What is seek time?

Q3. What does optical storage use to read data?

Q4. 1 byte contains how many bits?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

magnetic — Storage via magnetisation.

optical — Storage read with laser/light.

flash — Non-volatile electronic solid-state storage.

cost per GB — Price relative to capacity.

## New learning

Understand — Magnetic hard disks use magnetised regions on spinning platters with moving heads; they can offer large capacity at relatively low cost per GB.

Apply — Optical discs are read with a laser, often used for physically distributed or archived/removable media, though capacity and drive availability vary.

Evaluate — Flash uses electronic non-volatile memory cells without moving read/write parts; often faster random access and more robust to vibration than HDDs, but trade-offs remain.

## Worked example

Choose portable storage for a field photographer travelling over rough ground with active projects.

1. Explain the choice — Flash SSD: no spinning platter and generally quick random access; still requires backups.

# We Do · Practise with guidance

A school wants inexpensive bulk storage for a large local video archive, with separate backups.

Explain the choice

Magnetic HDD is a reasonable candidate because of capacity and cost per GB; validate resilience and transfer needs.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Match HDD, DVD and USB SSD to magnetic, optical and flash storage, with mechanisms. [6]

### Question 2

Explain one advantage and one limitation of flash SSD over magnetic HDD for a mobile video editor. [6]

### Question 3

Evaluate storage choice for distributing 100 offline revision packs to pupils and for a multi-terabyte editable school video archive. [8]

# Exit Ticket · Answer before checking

A hard disk drive reads bits using a laser.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Retains contents without power.

Q2. Time to position/locate data for a storage access.

Q3. Light/laser.

Q4. 8 bits.

### You Do pages · Independent questions

Q1. HDD magnetised platter [2], DVD laser optical disc [2], USB SSD flash electronic cells [2].

Q2. No moving parts → vibration resistance [2], fast random access [2], often higher cost/GB or write endurance trade-off [2].

Q3. Offline packs optical/removable flash depending hardware access [4]; HDD/SSD archive depending cost/speed [2]; backups and compatibility [2].

### Exit Ticket · Model answer

False. HDD uses magnetic read/write heads; optical discs use lasers.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**